



Amirkabir University of Technology
(Tehran Polytechnic)

Machine Learning Course By Dr. Nazerfard

CE5501 | Spring 2025

Teaching Assistants

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Assignment (2)

Outlines. In this assignment, some practical implementation skills which needed in this and other courses of this degree are noticed as well as linear regression and KNN topics. Remember that you may need to re-use your implementations of this assignment; so, it is suggested to code in functional.

Deadline. Please submit your answers before the end of date in courses.aut.ac.ir. Other methods like sending via email or in social networks are not accepted and will not be considered.

Assignment Manual

Delay policy. During the semester, you have extra 5 days for submitting your answers with delay. Mentioned time is for all assignments. After that, for each day of delay you lose 20% of the points of that assignment. After 4 days you miss all points and any submission will not be acceptable. Remember that saving this time doesn't have any extra point.

Sharing is not caring. Students are free to discuss and share their ideas about problems with others. But sharing source codes, solutions, answers and other results is not allowed and based on university's rule, both sides will be graded zero.

Problems are waiting for you. Some problems are required to be implemented within a programming language and obtain some charts, images, results, etc; then reasoned about. These types of questions are tagged by #Implementation. Some other problems are required to be solved or computed by hand or researched about. These types of questions are tagged by #Theoretical. You are not allowed to use programming language or other technical tools to answer theoretical problems.

Report is the key. All students' explanations, solutions, results, discussions and answers must be compacted into a single pdf report. A clean and explicit report is expected and may be followed by extra pts; so, you may need to write any related detail or experience during the solving problems. Report file should start within a cover page that includes course and assignment information as well as identical details like name, student number and email address. Second page should be a table of contents that indicates the student's answer to each question. Please repeat your name and student number on the left side of the footer in other pages. Also, you are free to write in Persian or English. If typing is bothering you, write in a paper and put its picture with acceptable readability in the report file.

Organize the upload items. Students should upload their implementation source codes as well as results and reports. You should upload a single .zip file with the following structure:

AML_02_[std-number].zip

Report

AML_02_[std-number].pdf [other material and results]

Source codes

P[problem-number]_[a-z].py

P[problem-number]_[a-z].ipynb

...

Python is the power. Students are free to use any programming language like python, matlab, C++ , etc. However it is strongly recommended to use python in the jupyter notebook environment; so, you may need to upload your .py or .ipynb sources.

Feel free to contact us. If you have any question or suggestion, need guidance or any comment be comfortable to ask via email as well as Telegram group.

Problem 1: Boston House | Regression (30 points)

In this exercise, you'll address a practical regression problem using the Boston House Prices dataset. Your goal is to predict the median value of owner-occupied homes in Boston suburbs using various features. This dataset, first introduced by Harrison and Rubinfeld (1978) and widely used in machine learning research, offers a valuable opportunity to hone your skills in data preprocessing, exploratory analysis, model construction, and evaluation.

Dataset Overview: The Boston House Prices dataset contains 506 observations with 13 predictive features and one target variable, which represents the median value of owner-occupied homes in \$1000's.

The Description of the dataset is taken from the below reference as shown in the table follows:

****Data Set Characteristics:****

:Number of Instances: 506

:Number of Attributes: 13 numeric/categorical predictive. Median Value (attribute 14) is usually the target.

:Attribute Information (in order):

- CRIM per capita crime rate by town
- ZN proportion of residential land zoned for lots over 25,000 sq.ft.
- INDUS proportion of non-retail business acres per town
- CHAS Charles River dummy variable (= 1 if tract bounds river; 0 otherwise)
- NOX nitric oxides concentration (parts per 10 million)
- RM average number of rooms per dwelling
- AGE proportion of owner-occupied units built prior to 1940
- DIS weighted distances to five Boston employment centres
- RAD index of accessibility to radial highways
- TAX full-value property-tax rate per \$10,000
- PTRATIO pupil-teacher ratio by town
- B $1000(Bk - 0.63)^2$ where Bk is the proportion of blacks by town
- LSTAT % lower status of the population

- 1) Load the Boston dataset using sklearn library and perform basic checks to verify the dataframe's dimensions and column names.
- 2) Is there any missing values in dataset? How would you handle missing values?
- 3) Determine the correlations between features, verify the shape of the correlation matrix, and visualize these relationships using a heatmap.
- 4) Select a feature for the prediction task and split the dataset into training and testing sets (test_size = 0.2,).

- 5) Import the Linear Regression, create a linear regressor, and train the model using the training dataset.
- 6) Retrieve the coefficient and intercept of the trained linear regression model to understand the relationship between the selected feature and the target variable.
- 7) Evaluate the performance of your linear regression model on the training data using the following metrics: R^2 , MAE, MSE, and RMSE. Explain these metrics.
- 8) Search and find how we can improve our regression model?
- 9) Visualize the differences between the actual prices and the predicted prices using a scatter plot.

Problem 2: Diabetes Prediction| KNN (40 points)

Type 2 diabetes is a chronic disease caused by insulin resistance, which, if left uncontrolled, can lead to heart, kidney, and nerve complications. Managing diabetes requires lifestyle changes, a healthy diet, and regular exercise. The main objective of this dataset is to predict whether a patient has diabetes or not. This dataset includes information on BMI, Insulin, SkinThickness, Pregnancies, Glucose, BloodPressure, DiabetesPedigree Function, Age, and Outcome. In this problem, we will use the KNN (K-Nearest Neighbors) algorithm to learn the feature weights and then select the best features for predicting and classifying individuals at risk of diabetes.



In **standard KNN (Vanilla KNN)**, the **Euclidean distance** is calculated as follows:

$$d(p,q) = \sqrt{\sum_{i=1}^n (q_i - p_i)^2}$$

In **Weighted KNN**, the distance is calculated as follows, where each feature **i** has a corresponding weight w_i :

$$d(A,B) = \sqrt{\sum_i w_i (A_i - B_i)^2}$$

In a standard KNN model, all features contribute equally to determining the similarity between samples. However, in the feature weighting method, certain features that have a greater impact on diabetes prediction are assigned higher weights.

- 1) Load the **diabetes.csv** dataset and perform the necessary preprocessing steps for the KNN algorithm. Explain these steps in your document.
- 2) Considering the importance of data balance in classification models, examine whether the data is balanced or not. If class imbalance exists, propose an appropriate solution to address it.

- 3) Use stratified sampling to split the data into 90% training set and 10% test set.
- 4) Calculate the weighted distance between each observation in the dataset and all other observations.
- 5) Use Cross-Validation to determine the optimal k for the KNN model. Train the model for different values of k and calculate the average accuracy to select the best k .
- 6) using the optimal k value and the weights obtained in the previous step, find the k nearest neighbors for each sample in the Test Set and predict its label based on the majority vote of its neighbors.
- 7) Initialize the feature weights to zero and then optimize them. After optimization, retrain the KNN model to minimize classification error.
- 8) For each sample in the test set, find the k nearest neighbors and predict its label based on the majority vote. Then, evaluate the model's performance and compare the results with the model before weight optimization.
- 9) Examine how an increase in features in the KNN model can lead to the Curse of Dimensionality and negatively impact the model's performance. To analyze this, compare two scenarios: (1) using all features in the model and (2) selecting features based on optimized weights. Run the model in both cases, evaluate its performance, and determine which approach yields better results.

Problem 3: Principal Component Analysis (PCA)| PCA (30points)

Step 1: Load and Visualize the Dataset:

Load the dataset using `load_digits` from `sklearn.datasets` and display 5 sample images.

- 1) How many features (dimensions) does each digit image have? What does each feature represent in the dataset?

Step 2: Apply PCA

Apply PCA to reduce dimensionality while retaining 70% of the variance.

- 2) How many principal components does PCA choose to retain 70% of the variance
- 3) Why is it better not to manually set the number of components (if we are not forced to)?
- 4) What would happen if we set `n_components=2` instead? How much variance would those two components capture?

Step 3: Visualizing PCA Components

Use PCA to project the data into two dimensions and visualize it using a scatter plot.

- 5) Does the scatter plot show a clear separation between different digits?
- 6) Could PCA alone be used for digit classification? Explain.

Step 4: Reconstruct Images from Reduced Dimensions

Reconstruct images using the compressed representation from PCA.

Display 5 original images alongside their reconstructed versions.

- 7) How would increasing the number of PCA components affect the reconstruction quality? Perform the reconstruction with a higher number of components and report your results.